Signals and Systems

Computer Assignment 3

Department of Electrical Engineering
Sharif University of Technology

Instructor: Dr. Hamid Behroozi

1. Algorithmic Implementation & Computational Complexity of the DFT

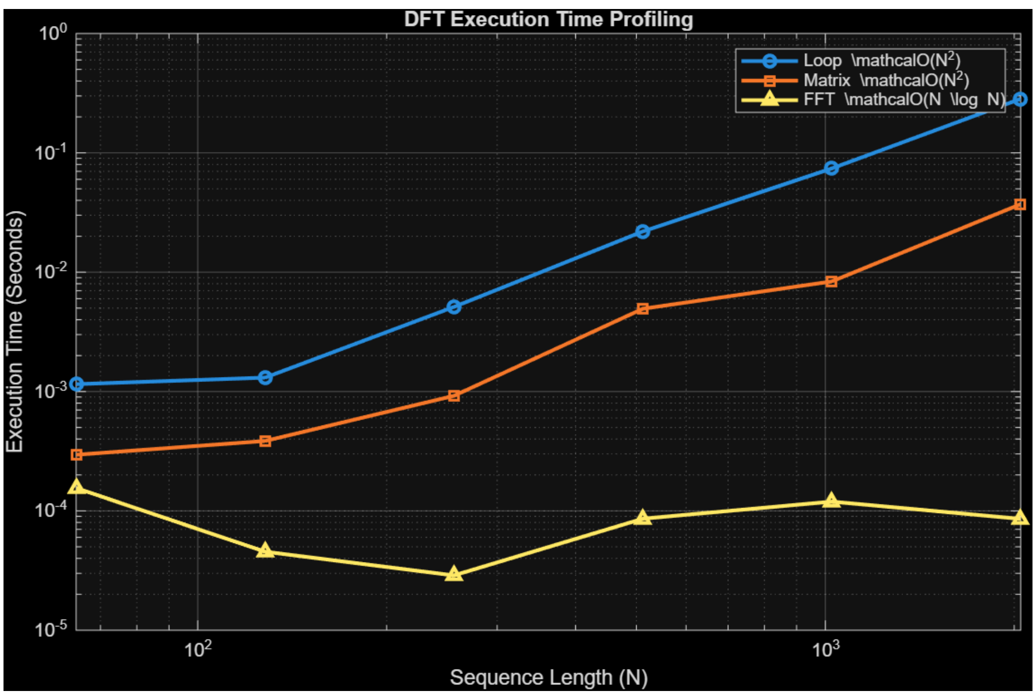
1.1. Direct Evaluation vs. Matrix Matrix Vectorization

```
function X = myDFTLoop(x)
    N = length(x);
    X = zeros(1, N);
    for k = 0:N-1
        for n = 0:N-1
            X(k+1) = X(k+1) + x(n+1) * exp(-1i * 2 * pi * k * n / N);
        end
    end
end

function X = myDFTMatrix(x)
    N = length(x);
    n = 0:N-1;
    k = 0:N-1;
    W = exp(-1i * 2 * pi / N * (k' * n));
    x_col = x(:);
    X_col = W * x_col;
    X = reshape(X_col, size(x));
end
```

1.2. Execution Profiling & Asymptotic Complexity

```
1.5429e-11

1.1177e-11
```

N	Loop_Time	Matrix_Time	FFT_Time
64	0.0011524	0.0002953	0.0001555
128	0.0013075	0.0003851	4.55e-05
256	0.0051158	0.0009205	2.88e-05
512	0.021951	0.0049406	8.6e-05
1024	0.073988	0.0083645	0.0001196
2048	0.28358	0.037146	8.57e-05

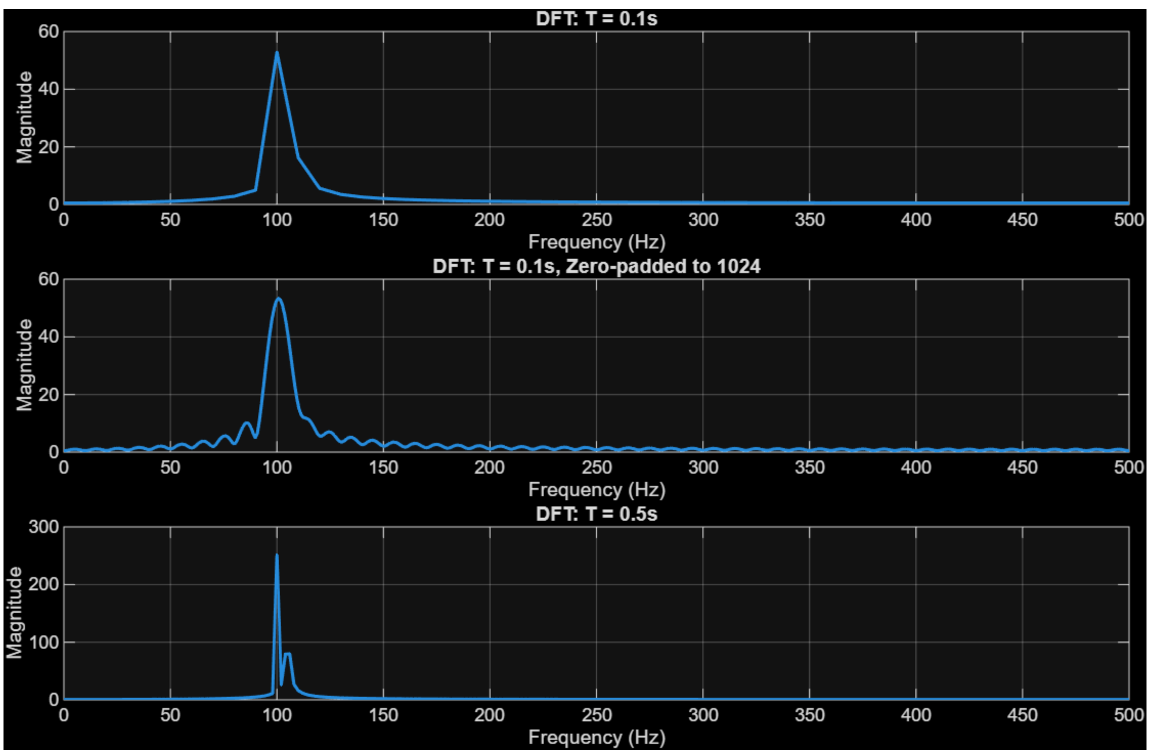


The nested loop approach and the matrix vectorization approach both possess a computational complexity of $\mathcal{O}(N^2)$. However, the execution speeds differ significantly; the matrix approach executes much faster due to the architectural benefits of memory vectorization. By eliminating explicit indexing loops in favor of a single matrix-vector multiplication, MATLAB can utilize highly optimized low-level linear algebra libraries.

Conversely, MATLAB's intrinsic `fft` function operates with an asymptotic complexity of $\mathcal{O}(N \log N)$. As sequence length N increases, the `fft` execution time scales logarithmically compared to the quadratic scaling of the custom functions, which is visually confirmed by the varying slopes in the log-log execution time plot.

2. Spectral Sampling, Frequency Resolution, and Leakage

2.1. The Effect of Signal Duration and Zero-Padding



(a)

For an observation window of $T = 0.1$ seconds , the fundamental frequency resolution is defined as $\Delta f = 1/T = 10$ Hz. Because the 5 Hz physical separation between the two sinusoidal components is smaller than the 10 Hz resolution, the two distinct sinusoidal peaks cannot be visually distinguished.

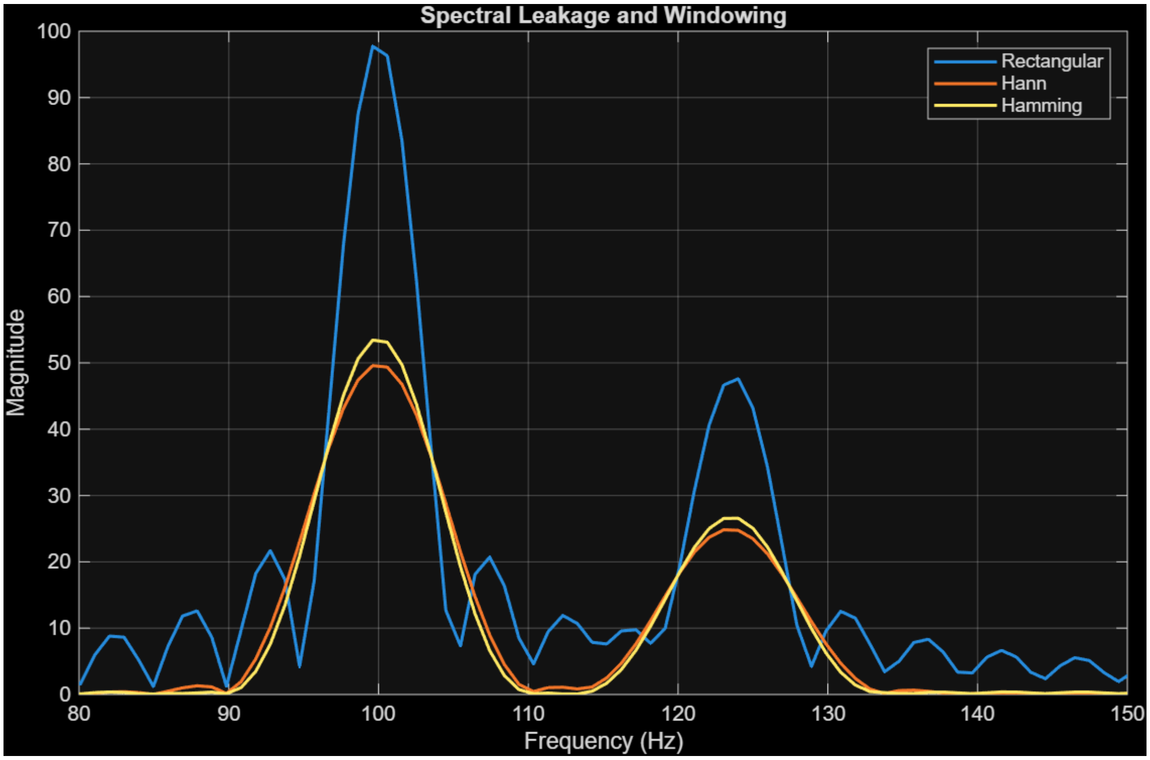
(b)

Padding the N_1 samples with zeros until $N_{pad} = 1024$ interpolates the DFT bins, resulting in a smoother DTFT envelope, but it does not improve the fundamental physical resolution of the spectrum.

(c)

Changing the observation window duration to $T = 0.5$ seconds without any zero-padding yields a resolution of $\Delta f = 2$ Hz. This successfully resolves the close spectral components, producing structural differences compared to the 0.1s window.

2.2. Windowing & Spectral Leakage



(d)

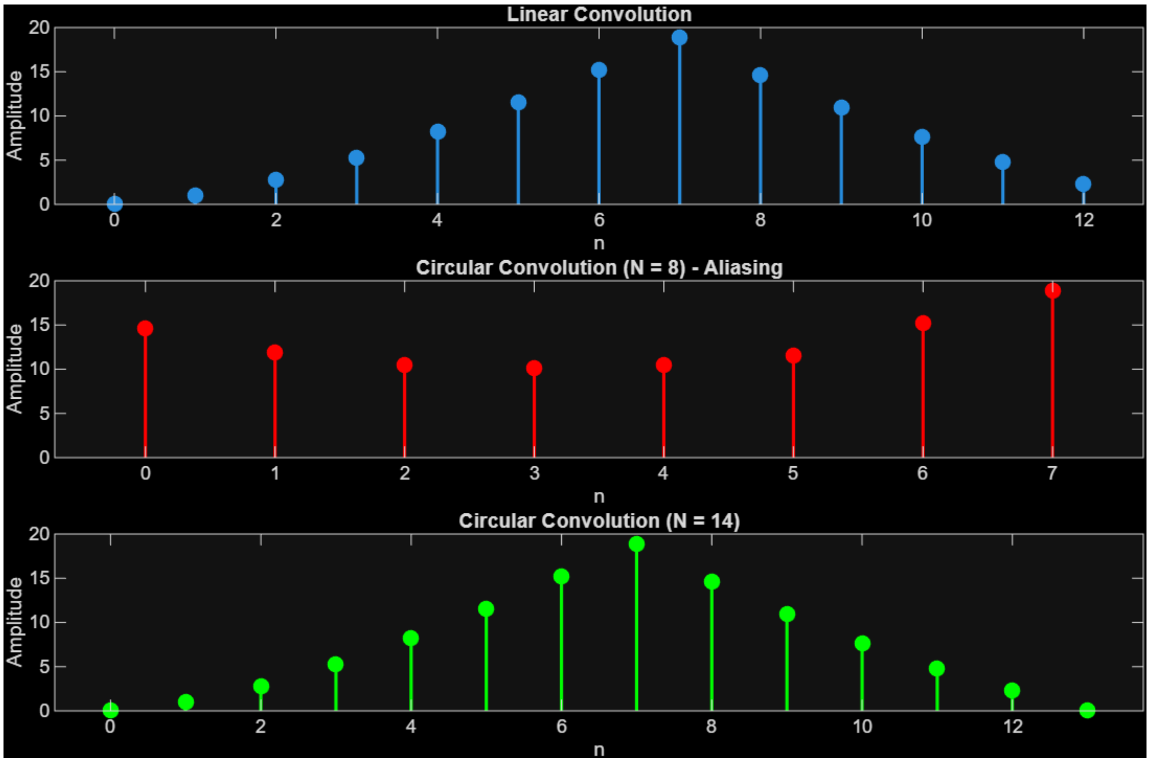
Modifying the signal parameters to $f_1 = 100$ Hz and $f_2 = 123.5$ Hz over a duration of $T = 0.2$ seconds creates a rectangular window truncation that does not align with a whole number of periods for f_2. This abrupt discontinuity manifests as "sloped skirts" surrounding the peaks, a phenomenon known as spectral leakage.

(f)

Applying tapered windows, such as a Hann window and a Hamming window, before calculating the DFT modifies the spectrum. There is a strict trade-off between a rectangular window and tapered windows. The rectangular window provides the narrowest main-lobe width (best spectral resolution), whereas tapered windows widen the main-lobe but provide exceptional side-lobe attenuation, which acts as a powerful mechanism for spectral leakage suppression.

3. Linear Convolution via the DFT (Circular Overlap)

3.1. Circular vs. Linear Convolution Mechanisms



(a)

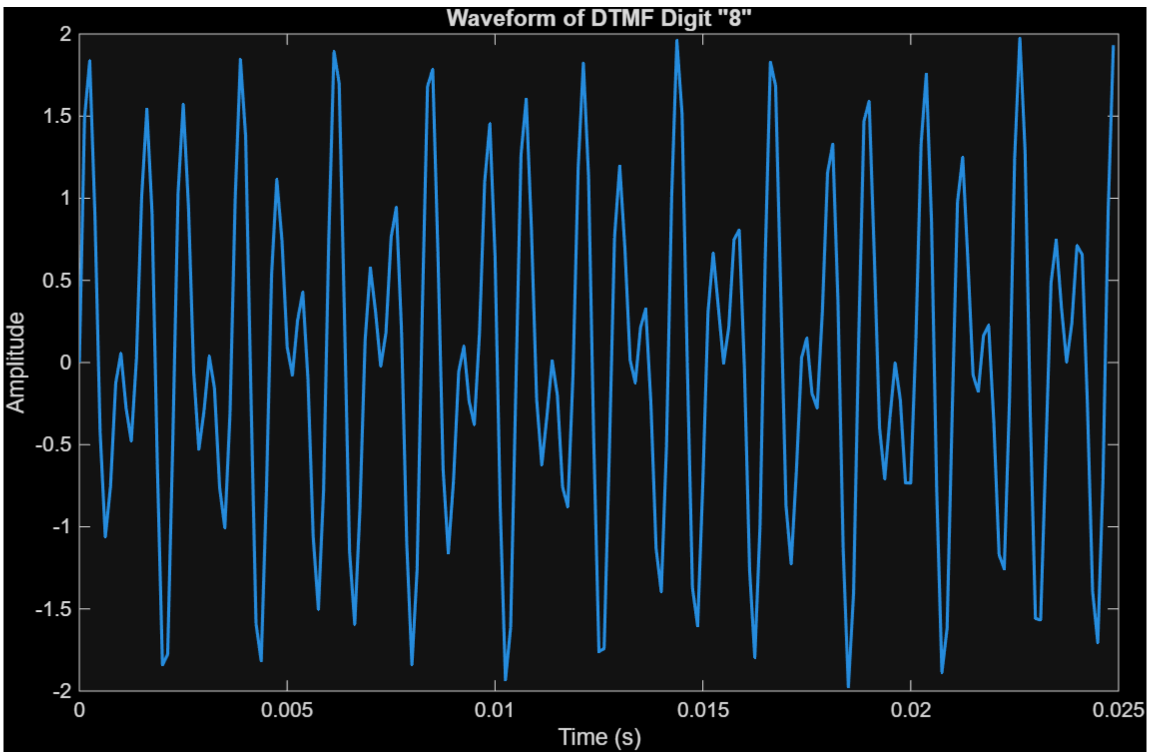
An N-point circular convolution of two finite sequences $x[n]$ (of length L) and $h[n]$ (oflength M) yields the exact same numerical values as their linear convolution if and only if $N \geq L + M - 1$.

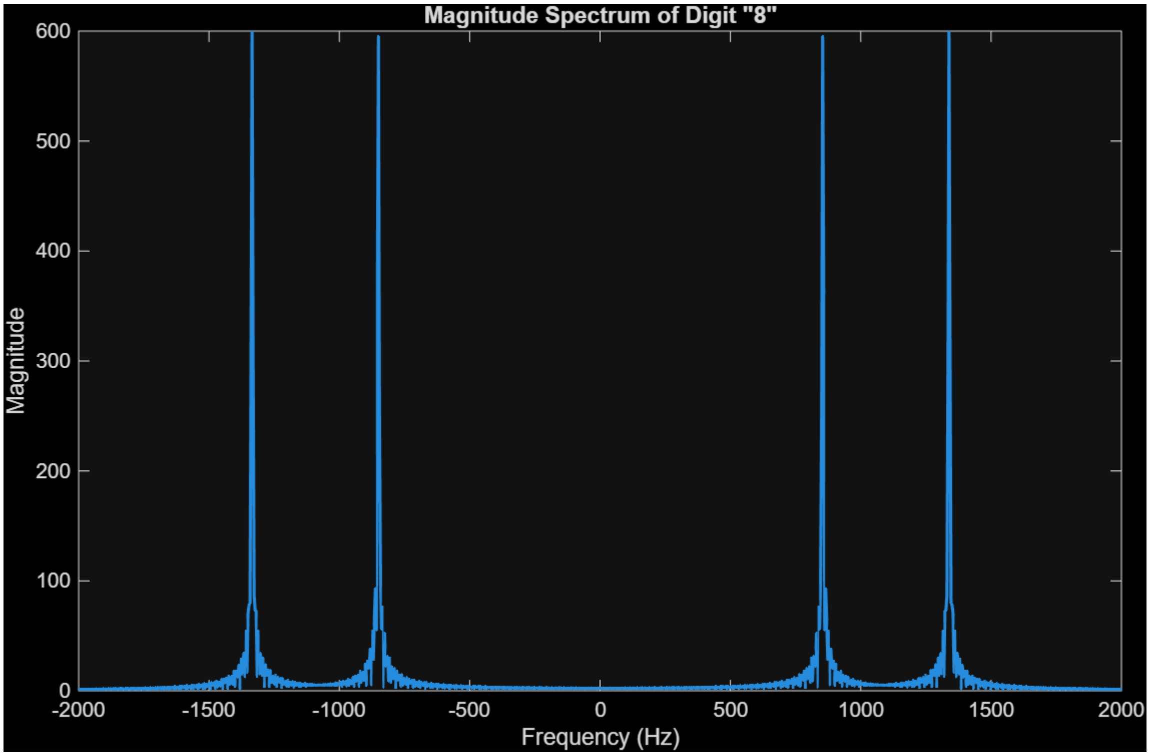
(d)

For the sequences defined in this experiment ($L = 8$ and $M = 6$), the minimum required length to prevent overlap is 13. When executing the operation via the `fft` and `ifft` commands for $N = 8$, time-domain aliasing occurs because the condition is violated. The tail of the linear convolution incorrectly wraps around and adds to the beginning of the sequence. For $N = 14$, the condition $14 \geq 13$ is satisfied, completely preventing aliasing and yielding numerical values identical to the baseline linear convolution.

4. Dual-Tone Multi-Frequency (DTMF) Touch-Tone Decoding

4.1. Signal Generation & Spectral Diagnostics



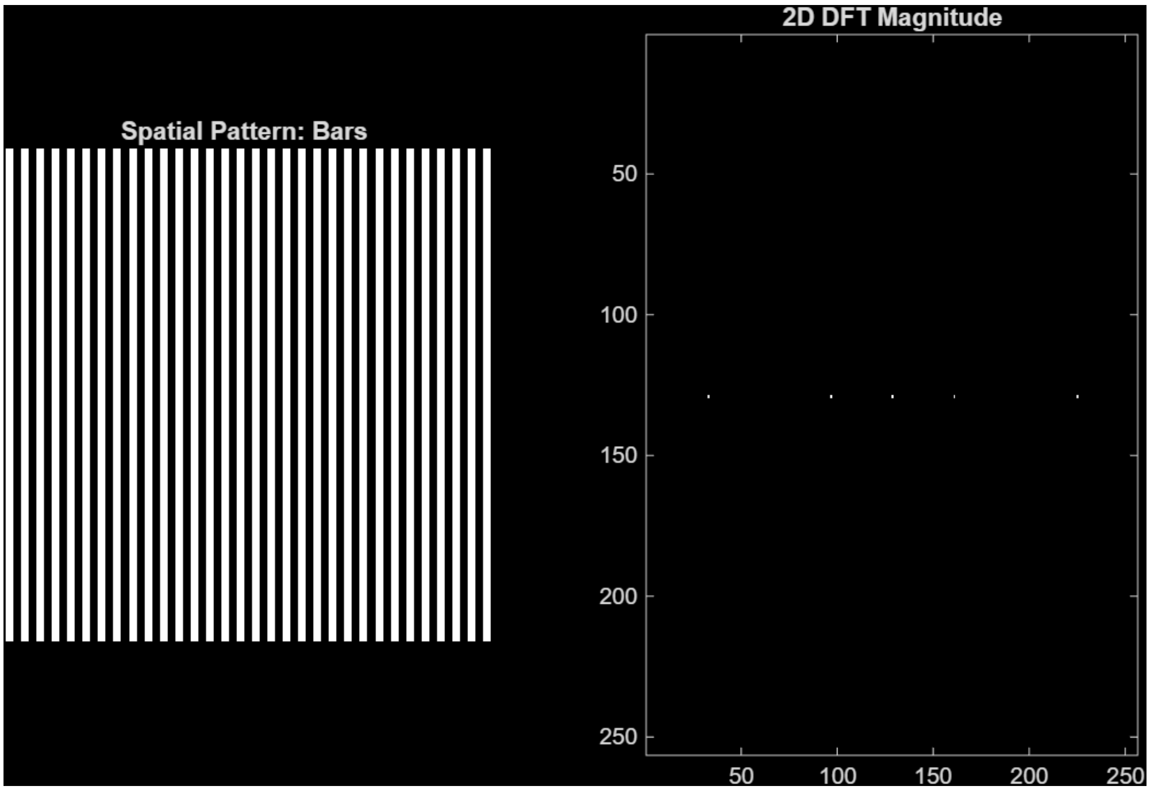


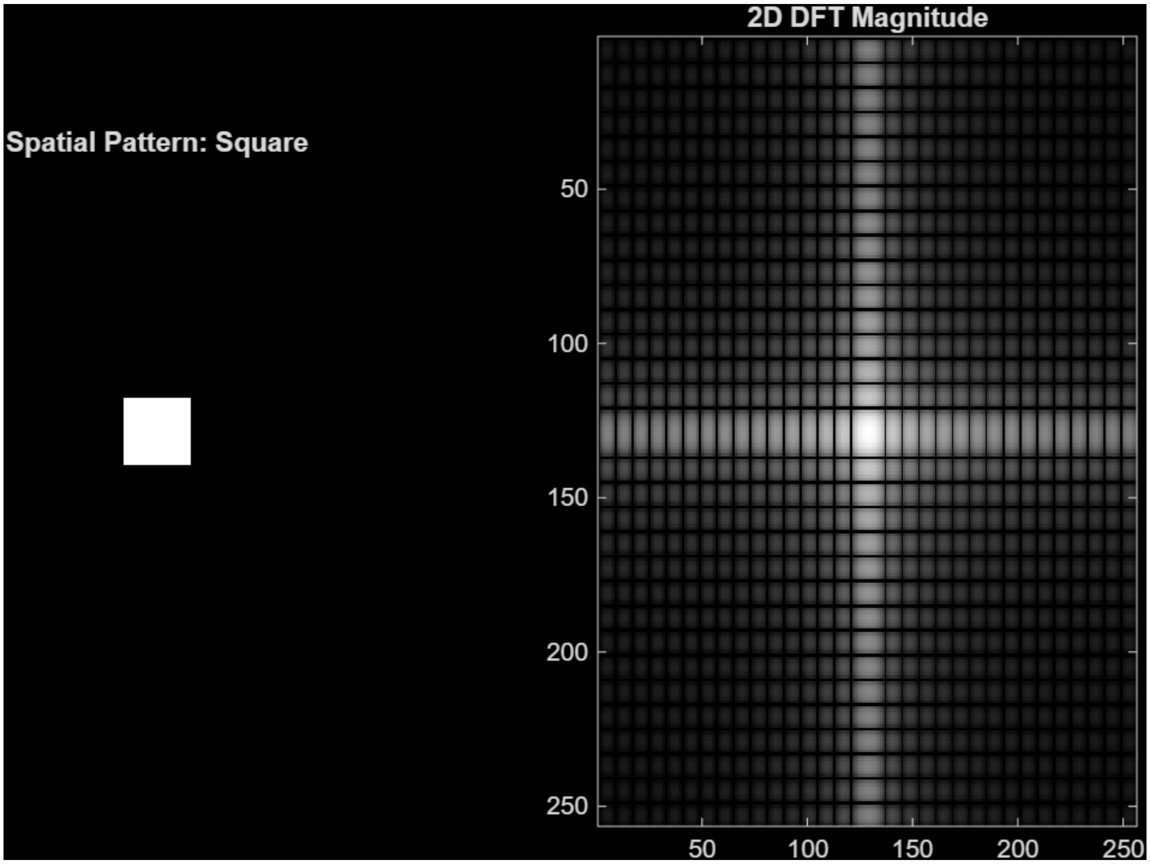
To synthesize the DTMF signal corresponding to dialing the digit ”8” , we combine its specific row and column frequencies. The synthesized time-domain waveform clearly displays the periodic characteristics resulting from the sum of these two sinusoids.

Upon computing the 2048-point DFT of the synthesized signal , the resulting magnitude spectrum provides a clear frequency-domain representation. We can verify that the prominent spectral peaks align exactly with the 852 Hz row and 1336 Hz column frequencies assigned to digit 8 in the standard specifications table.

5. Two-Dimensional DFT and Image Spectrogram Processing

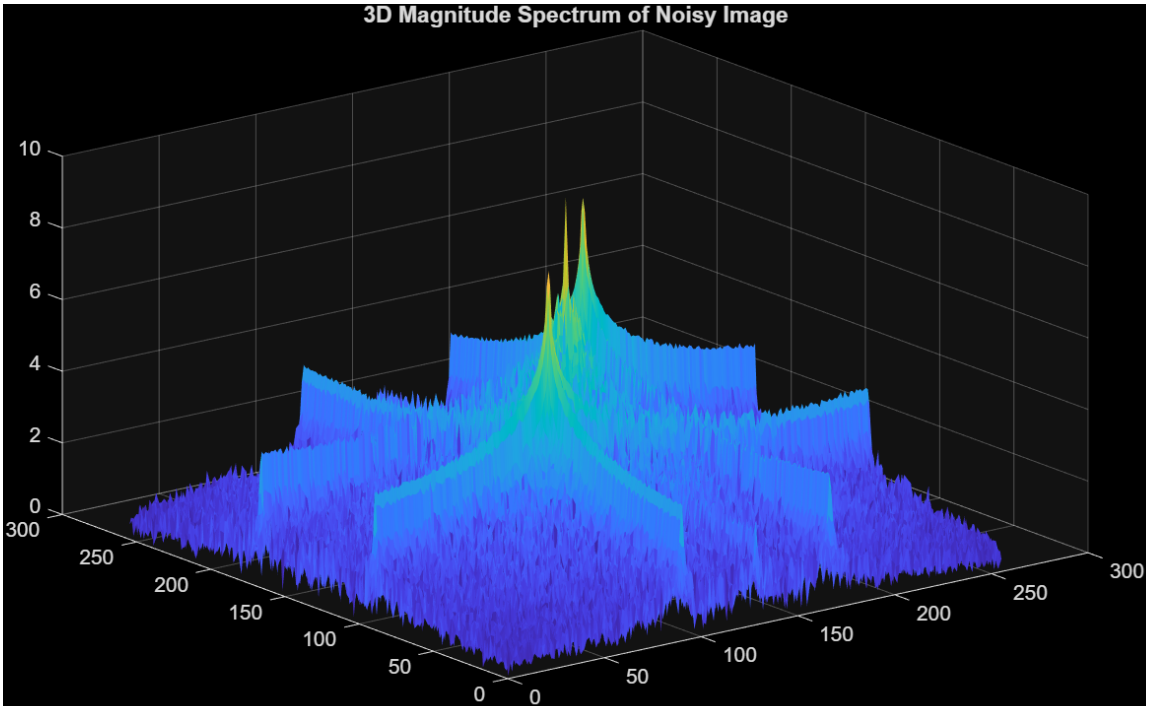
5.1. Spatial Patterns vs. 2D Spectral Topographies

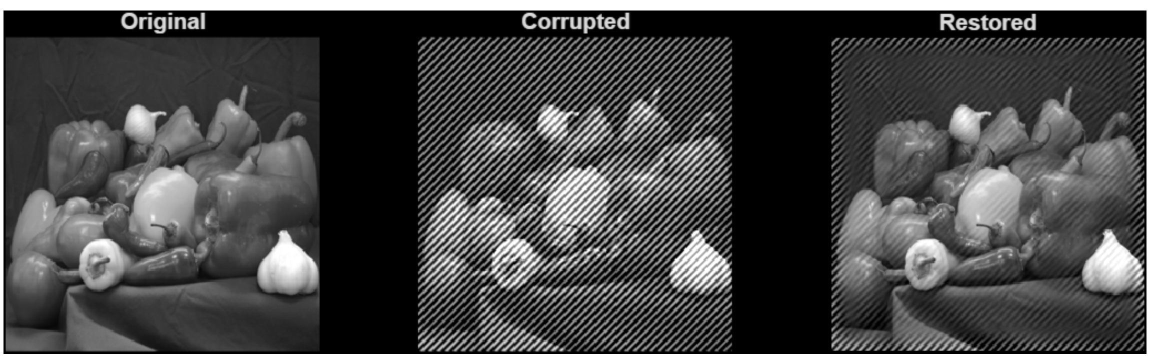




When transforming a binary image containing a solid centered square of size 32×32 pixels , the spatial square maps to a 2D cross-shaped sinc pattern in the frequency domain. This perfectly illustrates the inverse relationship between spatial dimensions and spectral bandwidth; compact spatial objects require high-frequency components to represent their sharp edges.

5.2. Frequency-Domain Structural Filtering





To remove periodic noise caused by a two-dimensional sinusoidal grid pattern , we design a 2D Frequency Notch Filter. We compute the centered 2D DFT of the corrupted image. The noise manifests as sharp, high-energy impulses in the magnitude spectrum.

By creating a masking matrix that equals 1 everywhere except for localized clusters of zeros centered directly over the coordinates of the noise impulses , and multiplying it element-wise with the shifted image spectrum, the noise is suppressed. After transforming the filtered spectrum back into the spatial domain, the image is successfully restored. Analyzing the performance of the filter shows that while edge preservation and image sharpness are largely maintained, setting a notch radius that is too large could introduce artificial ringing artifacts.